

**Harvard University
Curriculum Vitae**

Date Prepared: 09/09/2025

Name: Sara Nolin

Education:

05/2016	BA	Psychology	Samford University Birmingham, AL
06/2019	MA	Medical/Clinical Psychology (Advisor: Kristina Visscher)	University of Alabama at Birmingham Birmingham, AL
07/2022- 06/2023	Predoctoral Internship	Clinical Psychology: Neuropsychology track	Alpert Medical School of Brown University Providence, RI
08/2023	PhD	Medical/Clinical Psychology (Advisor: Kristina Visscher)	University of Alabama at Birmingham Birmingham, AL

Postdoctoral Training:

08/2023- 05/2025	Post-doctoral Fellowship	Neuropsychology (PI: Andreana Benitez)	Medical University of South Carolina Charleston, SC
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Committee Service:

Local

2017-2022	UAB Association of Neuropsychology Students & Trainees Interest Group	University of Alabama at Birmingham
2019-2021		Journal Club Chair
2022-2023	Psychology Training Committee	Alpert Medical School of Brown University
2022-2023		Intern Representative

National

2021-2024	KnowNeuropsychology Leadership Committee	KnowNeuropsychology
2021-2023		Volume Leader

Professional Societies:

2016-2018	Association for Psychological Science	
2017-2022	Organization for Human Brain Mapping	
2018-	Cognitive Neuroscience Society	
2018-2020	National Academy of Neuropsychology	
2016-	American Psychological Association	
2016-		Member, APA Division 40: Society for Clinical Neuropsychology
2019-2023		Member, APA Division 20: Adult Development and Aging
2021-2024		Member, Science Subcommittee of the American Association of Graduate Students
2021-2023		Member, Leadership Committee for the Special Interest Group on Coping, Resilience, and Successful Aging
2019	Society for Neuroscience	
2021-2023	International Neuropsychology Society	
2021-	Alzheimer's Association	
2024-	National Postdoctoral Association	
2024-2025		Member, Advocacy Committee

Editorial Activities:

- **Ad hoc Reviewer**

Brain Imaging and Behavior
Journal of the International Neuropsychology Society
Neuroimage: Clinical

- **Other Editorial Roles**

2021-2023	Associate Editor	Translational Issues in Psychological Sciences
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Honors and Prizes:

2016	Janice R. Teal Award	Samford University	classroom performance, research, service, career goals
2018	Abstract Competition Winner	University of Alabama at Birmingham	Annual Nathan Shock Symposium
2018	1 st place Award	University of Alabama at Birmingham	Comprehensive Center for Healthy Aging Symposium

2019	Runner-Up Poster	University of Alabama at Birmingham	Aging Symposium
2020	Runner –Up Presentation	University of Alabama at Birmingham	3-Minute Thesis Competition
2020,2021	Travel Award	University of Alabama at Birmingham Graduate Student Government	
2021	Travel Award	University of Alabama at Birmingham Civitan International Research Center	
2021	Travel Award	University of Alabama at Birmingham Department of Psychology	
2021	Edith Kaplan Scholarship	National Academy of Neuropsychology	Commitment to advancing neuropsychology through interdisciplinary collaboration
2023	Travel Award	Reserve and Resilience Workshop	
2024	Travel Award	South Carolina Alzheimer's Disease Research Center Symposium	
2024	Trailblazer Fellow	South Carolina Alzheimer's Disease Research Center's (SC-ADRC) Research Education Component	

Report of Funded and Unfunded Projects

Past

2019-2021	Relationship of cognitive performance to brain network dynamics in the healthy oldest old NIH T32 (NS061788-12 07/2008) UAB Cognition and Cognitive Disorders Institutional training grant Trainee This project sought to validate cognitive performance tests in the oldest-old and characterize their relationship with brain functional network organization.
2022-2023	Resting State Brain Network Dynamics with Electroencephalography (EEG) in middle-aged individuals with family history of Alzheimer's Disease Alpert Medical School of Brown University Predoctoral Seed Money Research Grant Co-PI (\$2,104.36) This project sought to evaluate network dynamics through EEG in middle-aged individuals with family history of Alzheimer's Disease as part of a larger study (PI: Laura Korthauer).

Report of Local Teaching and Training

Teaching of Students in Courses:

2018-2019	Methods in Neuroimaging Undergraduate	University of Alabama at Birmingham One hour lecture
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Research Supervisory and Training Responsibilities:

2016-2022	Training and supervision of research and thesis graduate, undergraduate, and high-school students (7 students)	University of Alabama at Birmingham 1:1 supervision one hour per week per student
2023-2025	Training and supervision of research and thesis 2 Undergraduate students	Medical University of South Carolina 1:1 supervision one hour per week per student

Report of Clinical Activities and Innovations**Past and Current Licensure and Board Certification:**

2025	Clinical Psychologist License, South Carolina
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Practice Activities:

2023-2025	Neuropsychological Assessment	Medical University of South Carolina	4 hours/week
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Clinical Innovations:

Developed and implemented group-based dementia caregiver intervention at the University of Alabama at Birmingham (2021-2022)	This dementia caregiver intervention filled a void in care to support caregiver's of patients with dementia that were being evaluated by the neuropsychology service locally at the University of Alabama at Birmingham. I was the lead developer with assistance from Kristine Lokken.
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Report of Education of Patients and Service to the Community

- ☒ *No presentations below were sponsored by 3rd parties/outside entities*
- ☐ *Those presentations below sponsored by outside entities are so noted and the sponsor(s) is (are) identified.*

Activities

2016-2017	University of Alabama at Birmingham Brain Bee / Presenter Gave presentations to high school students about various neuroscience topics
2017	GirlSpring / Presenter Gave presentation about training for a career in research

2017-2021 Dana Foundation / Brain Awareness Week presenter
 Gave presentations to K-12 children about various neuroscience topics

2022 Westminster High School, Birmingham, AL / Outreach presenter

Report of Scholarship

ORCID: 0000-0001-7107-1891

Peer-Reviewed Scholarship in print or other media:

Research Investigations

1. Robinson, J.-P., Anthony, T., Tripathi, R., **Sims, S.A.**, Visscher, K. M., & Bangalore, P. V. (2018). Analyzing the Human Connectome Project Datasets using GPUs. *PEARC '18: Practice and Experience in Advanced Research Computing*, 1–6.
2. **Sims, S.A.**, Demirayak, P., Cedotal, S., & Visscher, K. M. (2021). Frontal cortical regions associated with attention connect more strongly to central than peripheral V1. *NeuroImage*, 238, 118246. doi:10.1016/j.neuroimage.2021.118246
3. **Nolin, S.A.**, Cowart, H., Raichlen, D.A., Franchetti, M.K., Rezaei, R.F., Merritt, S., Jessup, C.J., Geldmacher, D., Hishaw, G.A., Alperin, N., Trouard, T.P., Wadley, V.G., Levin, B.E., Woods, A.J., Rundek, T., Cohen, R.A., Alexander, G.E., Visscher, K.M. (2022) Validity of the NIH Toolbox Cognitive Battery in a healthy oldest-old 85+ sample. *Journal of the International Neuropsychology Society*, 1-10. doi:10.1017/S1355617722000443.
4. Norling, M., Bennett, A., Crowe, M., Long, D.L., **Nolin, S.A.**, Myers, T., Del Bene, V.A., Lazar, R.M., Gerstenecker, A. (2023) Longitudinal Effects of Anticholinergic Medications on Cognition and Likely Mitigating Effects of Physical Activity. *Journal of the American Geriatrics Society*, 71(6):1937-1943. doi: 10.1111/jgs.18279
5. Britton, M., Jensen, G., Edden, R., Puts, N., **Nolin, S.A.**, Merritt, S., Rezaei, R., Forbes, M., Johnson, K., Bharadwaj, P., Franchetti, M., Raichlen, D., Jessup, C., Hishaw, G., Van Etten, E., Gudmundson, A., Murali-Manohar, S., Cowart, H., Trouard, T., Geldmacher, D.S., Wadley, V.G., Alperin, N., Levin, B.E., Rundek, T., Visscher, K.M., Woods, A.J., Alexander, G.E., Cohen, R.A., Porges, E. (2024) “Surviving and Thriving”: Evidence for Cortical GABA Stabilization in Cognitively-Intact Oldest-Old Adults. *Translational Psychiatry*, 15(79). doi: 10.1038/s41398-025-03302-w
6. **Nolin, S.A.**, Faulkner, M.E., Stewart, P., Raichlen, D.A., Bharadwaj, P.K., Franchetti, M.K., Rezaei, R.F., Merritt, S., Jessup, C.J., Porges, E.S., Geldmacher, D., Hishaw, G.A., Alperin, N., Trouard, T.P., Wadley, V.G., Levin, B.E., Woods, A.J., Rundek, T., Cohen, R.A., Alexander, G.E., Visscher, K.M. (2025) Network Segregation Predicts Processing Speed in the Cognitively Healthy Oldest-old. *eLife*. 14:e78076. doi:10.7554/eLife.78076
7. Knox, S., Aghamoosa, S., Cutty, M., Wright, A., Scherbakov, D., Hood, G., **Nolin, S.A.**, Obeid, J. (2025) AI Approaches for Phenotyping Alzheimer’s Disease and Related Dementias Using Electronic Health Records. *Alzheimer’s & Dementia: Translational Research & Clinical Interventions*. 11(2):e70089. doi:1.1002/trc2.70089
8. Aghamoosa, S., **Nolin, S.A.**, Chen, A., Caulfield, K.A., Lopez, J., Rbeiz, K., Fleischmann, H.H., Horn, O., Antonucci, M., Revuelta, G., McTeague, L.M., Benitez, A. (2025) iTBS-Induced Changes in Resting-State Functional Connectivity Relate to Cognitive Improvement in MCI. *Brain Stimulation*. 18(3):957-964. doi: 10.1016/j.brs.2025.04.012

Other peer-reviewed scholarship

1. McDonough, I., **Nolin, S.A.**, Visscher, K.M. (2022) 25 Years of Neurocognitive Aging Theories: What Have We Learned? *Frontiers in Aging Neuroscience*. 14:1002096. doi: 10.3389/fnagi.2022.1002096

Manuscripts Submitted to Preprint Servers

1. Parupura, V., Caceres, H., **Nolin, S.A.**, Raichlen, D.A., Franchetti, M.K., Rezaei, R.F., Merritt, S., Jessup, C.J., Geldmacher, D., Hishaw, G.A., Alperin, N., Trouard, T.P., Wadley, V.G., Levin, B.E., Woods, A.J., Rundek, T., Cohen, R.A., Alexander, G.E., Visscher, K.M. The relationship between hippocampal structure and episodic memory in the oldest-old. bioRxiv 2022.04.10.487750; doi: 10.1101/2022.04.10.487750

Abstracts, Poster Presentations, and Exhibits Presented at Professional Meetings:

1. **Nolin, S.A.**, Faulkner, M.E., Stewart, P., Raichlen, D.A., Bharadwaj, P.K., Franchetti, M.K., Rezaei, R.F., Merritt, S., Jessup, C.J., Porges, E.S., Geldmacher, D., Hishaw, G.A., Alperin, N., Trouard, T.P., Wadley, V.G., Levin, B.E., Woods, A.J., Rundek, T., Cohen, R.A., Alexander, G.E., Visscher, K.M. (2023, February) Fronto-parietal Network Segregation Predicts Maintained Cognition in the Cognitively Healthy Oldest-old (85+): evidence for dedifferentiation. Poster presentation at INS, San Diego, CA.
2. **Nolin, S.A.**, Gemelii, Z., Korthauer, L. (2023, March) Relationship between Cognition and EEG Resting State Brain Network Organization in Healthy Older Adults. Poster presentation at Mind Brain Research Day, Brown University Department of Psychiatry, Providence, RI.
3. Aghamoosa, S.*, **Nolin, S.A.***, Chen, A.A., Caulfield, K.A., Lopez, J., Rbeiz, K., Fleischmann, H.H., Horn, O., Madden, K., Antonucci, M., Revuelta, G., McTeague, L.M., & Benitez, A. (2024, July). Increased functional connectivity relates to cognitive improvement following non-invasive brain stimulation in MCI. Poster Presentation at Alzheimer's Association International Conference 2024, Philadelphia, PA.
4. Ling, G., Cowart, H., **Nolin, S.A.**, Stewart, P., Fleming, L., Faulkner, M.E., Merritt, S., Rezaei, R.F., Bharadwaj, P.K., Franchetti, M.K., Raichlen, D.A., Jessup, C.J., Hishaw, G.A., Van Etten, E.J., Trouard, T.P., Geldmacher, D., Wadley, V.G., Alperin, N., Porges, E.C., Woods, A.J., Cohen, R.A., Levin, B.E., Rundek, T., Alexander, G.E., Visscher, K.M. (2024). Dynamics of Dominant Default Mode Network Linked to Processing Speed in Cognitively Healthy Oldest-Old. Poster Presentation at the Society for Neuroscience, Chicago, IL.
5. Knox, S., Aghamoosa, S., Cutty, M., Wright, A., Scherbakov, D., Hood, G., **Nolin, S.A.**, Obeid, J. (2024). Automated Identification of Patients with Alzheimer's Disease and Related Dementias in Medical Records using AI. Oral Presentation at the South Carolina ADRC Symposium, Greenville, SC.
6. **Nolin, S.A.**, Aghamoosa, S., Jones, A., Jensen, J., Falangola, M. F., Rodriguez-Porcel, F., Spampinato, M. V., Rieter, W.J., Hamlett, E., Nietert P., Madden, K., Helton, S., Thorn, K., Oh, H., Benitez, A. (2024). Investigating amyloid deposition across brain functional networks in preclinical AD. Poster Presentation at the South Carolina ADRC Symposium, Greenville, SC.
7. **Nolin, S.A.**, Aghamoosa, S., Jones, A., Nietert P., Benitez, A. (2025) Greater Amyloid Accumulation in Cognitive Networks in Preclinical Alzheimer's Disease. Poster Presentation at Cognitive Neuroscience Society Annual Meeting, Boston, MA.

Narrative Report

My research focuses on how brain networks support cognition in aging and Alzheimer's disease, with the long-term goal of identifying early markers of decline and resilience. During my doctoral training at the University of Alabama at Birmingham, I investigated cognition and brain network organization in the oldest-old (85+). I demonstrated that preserved cognition in this group is associated with more

differentiated network organization, contributing to our understanding of mechanisms that support successful aging.

As a predoctoral intern at Brown University, I expanded my expertise to include electroencephalography (EEG) and led a funded project examining resting-state network dynamics in middle-aged adults with a family history of Alzheimer's disease. This experience shaped my interest in dynamic connectivity methods as sensitive markers of early brain change. As a postdoctoral fellow at the Medical University of South Carolina, I extended this work by integrating positron emission tomography (PET) with functional connectivity to examine how amyloid deposition interacts with cognitive networks in preclinical Alzheimer's disease. These projects reflect my commitment to advancing multimodal approaches to the study of brain aging.

My clinical training as a neuropsychologist informs my research by grounding scientific questions in patient care. I have conducted comprehensive neuropsychological assessments and co-developed a dementia caregiver intervention to address unmet needs in clinical services. I have also been dedicated to teaching and mentorship, supervising undergraduate, graduate, and high school trainees, and contributing nationally through leadership roles in professional societies and editorial activities.

Moving forward, I aim to build a program of research that uses advanced imaging and analytic approaches to characterize brain network dynamics across the spectrum of aging and Alzheimer's disease. By linking connectivity measures with molecular pathology and cognition, my goal is to contribute to the development of sensitive, clinically relevant biomarkers that can inform early intervention and prevention strategies.